Problem H3: Calculating the Weak Coupling Constant g_W Two Ways

PHYS 7502 — Particles and Nuclei 2

Problem Statement (H3)

Calculate the weak coupling constant g_W two ways:

1. Using Thomson Eq. 17.39 (from the handout) and the Higgs vacuum expectation value v (also given elsewhere in the Thomson handout), along with the mass of the W boson.
2. Using Griffiths Equation 8.26, along with the weak mixing angle relations in Griffiths Equation 9.91 and equations around it.

Compare the two results.

Deliverables

- A numerical value of g_W from **Method 1** (Higgs mechanism / Thomson).
- A numerical value of g_W from **Method 2** (Electroweak mixing / Griffiths).
- A comparison of the two results, showing they are consistent.

1 Background Context: What is g_W and Why Two Methods?

Physical Intuition: Why This Problem Matters

The weak coupling constant g_W is one of the fundamental parameters of the Standard Model. It sets the strength of the $SU(2)_L$ gauge interaction — the charged weak force mediated by $W^{\pm}$ bosons. This problem asks you to extract g_W from two *completely independent* sets of relationships, demonstrating the internal consistency of the electroweak theory:

- **Method 1** comes from the **Higgs mechanism**: how the W boson acquires its mass through spontaneous symmetry breaking.
- **Method 2** comes from the **electroweak unification**: the relationship between the electromagnetic coupling, the weak coupling, and the weak mixing angle.

Getting the *same answer* from two independent routes is a powerful consistency check on the entire Glashow–Salam–Weinberg (GSW) model.

Key Concepts Needed

The Higgs Mechanism (Thomson Chapter 17): When the Higgs field acquires a non-zero vacuum expectation value v, the W and Z bosons gain mass. The W-boson mass is directly proportional to g_W and v.

Electroweak Unification (Griffiths Chapter 9): The GWS model unifies the electromagnetic interaction (coupling g_e) with the weak interaction (coupling g_W) through the weak mixing angle θ_W. The electromagnetic coupling constant $g_e = \sqrt{4\pi\alpha}$ is related to g_W via $\sin\theta_W$.

2 Part 1: g_W from the Higgs Mechanism (Thomson Eq. 17.39)

Restated Question — Part 1

Using the Thomson relation $m_W = \frac{1}{2} g_W v$ (Eq. 17.39) and the known values of the W-boson mass m_W and the Higgs vacuum expectation value v, solve for g_W.

Deliverables — Part 1

- A numerical value of g_W extracted from the Higgs mechanism relation.

2.1 Source References

Where to Find These Equations

- **Thomson Eq. 17.39**: Section 17.5 ("The Higgs Mechanism"), p. 481 — derives the W-boson mass from the covariant derivative acting on the Higgs doublet.
- **Higgs VEV** $v = 246$ GeV: Thomson Section 17.5, p. 483 (also Griffiths Eq. 12.1 in the revised edition gives $v = 2M_W c^2/g_W = 246$ GeV).
- **W-boson mass**: $m_W = 80.379\ \text{GeV}/c^2$ (PDG value; Thomson uses $m_W = 80.385\ \text{GeV}/c^2$).

2.2 Background: Where Thomson Eq. 17.39 Comes From

Term Refresher: Vacuum Expectation Value (VEV)

In the Standard Model, the Higgs field is a complex $SU(2)_L$ doublet. The "vacuum expectation value" v is the value of the Higgs field in the lowest-energy state (the vacuum). Because the Higgs potential has the "Mexican hat" shape with $\mu^2 < 0$, the minimum of the potential does *not* occur at $\phi = 0$, but rather at a non-zero value:

$$v = \sqrt{\frac{-\mu^2}{\lambda}}$$

where μ^2 and λ are parameters in the Higgs potential $V(\phi) = \mu^2|\phi|^2 + \lambda|\phi|^4$.

The Higgs doublet in the unitary gauge takes the form:

$$\phi = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}$$

where $h(x)$ is the physical Higgs boson field (the excitation around the vacuum).

When we compute the kinetic term $(D_\mu \phi)^\dagger (D^\mu \phi)$ using the $SU(2)_L \times U(1)_Y$ covariant derivative, the terms quadratic in the $W^{(1)}$ and $W^{(2)}$ gauge fields are:

$$\frac{1}{8} v^2 g_W^2 \left[W_\mu^{(1)} W^{(1)\mu} + W_\mu^{(2)} W^{(2)\mu} \right]$$

In the Lagrangian, a massive spin-1 field V_μ has a mass term of the form $\frac{1}{2}m_V^2\, V_\mu V^\mu$. Comparing:

$$\frac{1}{2}\, m_W^2\, W_\mu W^\mu = \frac{1}{8}\, g_W^2\, v^2\, W_\mu W^\mu$$

Thomson Equation 17.39

$$\boxed{m_W = \frac{1}{2}\, g_W\, v}$$

The W-boson mass equals one-half the weak coupling constant times the Higgs VEV.

2.3 Strategy for Part 1

- **Step 1**: Write down Thomson Eq. 17.39: $m_W = \frac{1}{2} g_W v$.
- **Step 2**: Solve algebraically for g_W.
- **Step 3**: Plug in the known numerical values of m_W and v.
- **Step 4**: Compute the numerical answer.

2.4 Solution

Starting from Thomson Eq. 17.39:

$$m_W = \frac{1}{2}\, g_W\, v$$

Solve for g_W:

$$g_W = \frac{2\, m_W}{v}$$

Numerical Inputs

From Thomson (and PDG):

$$m_W = 80.379 \text{ GeV}/c^2, \qquad v = 246 \text{ GeV}$$

(Note: Thomson uses $m_W = 80.385$ GeV/c^2; we use the rounded PDG value. The difference is negligible at this level.)

Substituting:

$$g_W = \frac{2 \times 80.379 \text{ GeV}}{246 \text{ GeV}}$$

$$g_W = \frac{160.758}{246}$$

Result — Part 1

$$\boxed{g_W \approx 0.653}$$

Physical Insight: What This Number Means

The "weak fine structure constant" is:

$$\alpha_W = \frac{g_W^2}{4\pi} = \frac{(0.653)^2}{4\pi} \approx \frac{0.427}{12.57} \approx 0.034 \approx \frac{1}{29}$$

This is *larger* than the electromagnetic fine structure constant $\alpha \approx 1/137$! The weak interaction is intrinsically *stronger* than electromagnetism. It only appears "weak" at low energies because the W and Z propagators suppress the amplitude by a factor of $\sim q^2/m_W^2$ when $|q^2| \ll m_W^2$.

3 Part 2: g_W from Electroweak Mixing (Griffiths Eqs. 8.26 & 9.91)

Restated Question — Part 2

Using the electromagnetic coupling constant g_e (related to the fine structure constant α, Griffiths Eq. 8.26) and the weak mixing angle relation $g_W = g_e/\sin\theta_W$ (Griffiths Eq. 9.91), calculate g_W.

Deliverables — Part 2

- A numerical value of g_W extracted from the electroweak unification relations.

3.1 Source References

Where to Find These Equations

- **Griffiths Eq. 8.26**: Chapter 8 (Electrodynamics and Chromodynamics of Quarks) — defines the QED coupling constant: $g_e = \sqrt{4\pi\alpha}$, where $\alpha \approx 1/137.036$ is the fine structure constant.
- **Griffiths Eq. 9.91**: Chapter 9 (Weak Interactions), Section 9.6 — the weak mixing angle relations:
$$g_W = \frac{g_e}{\sin\theta_W}, \qquad g_Z = \frac{g_e}{\sin\theta_W \cos\theta_W}$$
- **Griffiths Eq. 9.92**: $M_W = M_Z \cos\theta_W$ (mass relation).
- **Griffiths Eq. 9.93**: $\sin^2\theta_W = 0.2314$ (experimental value of the weak mixing angle).

3.2 Background: The Electromagnetic Coupling Constant g_e

Term Refresher: Fine Structure Constant α

The fine structure constant is the fundamental dimensionless parameter of QED:

$$\alpha = \frac{e^2}{4\pi\,\epsilon_0\,\hbar c} \approx \frac{1}{137.036}$$

In Griffiths' notation (Gaussian units with $\hbar$ and c explicit), the QED vertex factor at each electromagnetic vertex is $ig_e\gamma^\mu$, where the dimensionless coupling constant g_e is defined by:

$$g_e = \sqrt{4\pi\alpha}$$

This is sometimes written $g_e = e/\sqrt{\hbar c}$ in Gaussian units. In Heaviside–Lorentz natural units ($\hbar = c = 1$), g_e is simply the positron charge e, and $\alpha = e^2/(4\pi)$.

Convention Warning

Different textbooks define "e" differently depending on their unit system. In Griffiths (Gaussian), $g_e = \sqrt{4\pi\alpha}$. In Thomson and many modern texts (Heaviside–Lorentz natural units), $e = \sqrt{4\pi\alpha}$. The physics is the same; only the labeling differs. What matters is the *dimensionless* number $\alpha = 1/137.036$.

3.3 Background: The Weak Mixing Angle

Term Refresher: Weak Mixing Angle θ_W

In the GWS electroweak theory, the physical photon A^μ and Z^0 boson are *mixtures* of the underlying $SU(2)_L$ gauge boson $W^{(3)\mu}$ and the $U(1)_Y$ gauge boson B^μ:

$$A^\mu = B^\mu \cos\theta_W + W^{(3)\mu} \sin\theta_W$$

$$Z^\mu = -B^\mu \sin\theta_W + W^{(3)\mu} \cos\theta_W$$

The mixing angle θ_W (also called the Weinberg angle) parameterizes this rotation. Consistency of the unified electroweak theory with ordinary QED requires:

$$g_W \sin\theta_W = g' \cos\theta_W = g_e$$

where g' is the $U(1)_Y$ coupling constant.

Why This Works

The electromagnetic coupling g_e is extremely well-measured (via the fine structure constant). The weak mixing angle $\sin^2\theta_W$ is also precisely measured (e.g., from Z-pole measurements at LEP). So we can extract g_W without ever needing the Higgs VEV or the Fermi constant — a completely independent route from Part 1.

3.4 Strategy for Part 2

- **Step 1**: Compute $g_e = \sqrt{4\pi\alpha}$ from the known fine structure constant (Griffiths Eq. 8.26).
- **Step 2**: Use $\sin^2\theta_W = 0.2314$ to get $\sin\theta_W$ (Griffiths Eq. 9.93).
- **Step 3**: Apply $g_W = g_e/\sin\theta_W$ (Griffiths Eq. 9.91).
- **Step 4**: Compute the numerical answer.

Alternative strategy: One could also go through the Fermi constant G_F and m_W using the relation $G_F/(\hbar c)^3 = \frac{\sqrt{2}}{8}(g_W/M_W c^2)^2$ (Griffiths Eq. 9.36). This gives yet another independent check, but the problem specifically asks for the electroweak mixing approach.

3.5 Solution

Step 1: Compute g_e

The electromagnetic coupling constant is:

$$g_e = \sqrt{4\pi\alpha} \tag{1}$$

Using $\alpha = 1/137.036$:

$$g_e = \sqrt{\frac{4\pi}{137.036}} \tag{2}$$

$$= \sqrt{\frac{12.566}{137.036}} \tag{3}$$

$$= \sqrt{0.09170} \tag{4}$$

Intermediate Result: g_e

$$g_e = 0.3028$$

Step 2: Compute $\sin\theta_W$

From Griffiths Eq. 9.93, the experimental value of the weak mixing angle is:

$$\sin^2\theta_W = 0.2314$$

Taking the square root:

$$\sin\theta_W = \sqrt{0.2314} = 0.4810$$

This corresponds to $\theta_W \approx 28.8°$.

Pitfall: Degrees vs. Radians

Always be clear about whether you are working in degrees or radians. Here we only need $\sin\theta_W$ (a dimensionless number), so the angle unit does not matter for the final computation. But if you were to state the angle itself, it is 28.8° or 0.502 rad.

Step 3: Apply $g_W = g_e/\sin\theta_W$

From Griffiths Eq. 9.91:

$$g_W = \frac{g_e}{\sin\theta_W}$$

Substituting our computed values:

$$g_W = \frac{0.3028}{0.4810}$$

Result — Part 2

$$\boxed{g_W \approx 0.630}$$

Why the Slight Difference?

This value of $g_W \approx 0.630$ is close to, but not exactly equal to, the Part 1 result of $g_W \approx 0.653$. This small discrepancy arises because the Fermi constant G_F, the W mass, α, and $\sin^2\theta_W$ each receive radiative corrections (loop-level quantum corrections). At tree level, both methods give the *same* g_W. The $\sim$3% difference reflects the fact that we are comparing tree-level relations using experimentally measured quantities that already include radiative corrections. In particular, $\sin^2\theta_W = 0.2314$ is the *effective* value measured at the Z pole, which differs slightly from the tree-level prediction. Using the on-shell definition $\sin^2\theta_W = 1 - m_W^2/m_Z^2$, one gets $\sin^2\theta_W \approx 0.2229$, which brings the two methods into closer agreement.

4 Comparison of the Two Methods

Method	Key Equation	Inputs Used	Result for g_W
1: Higgs VEV	$m_W = \frac{1}{2} g_W v$	$m_W = 80.4$ GeV, $v = 246$ GeV	≈ 0.653
2: Electroweak Mixing	$g_W = g_e / \sin\theta_W$	$\alpha = 1/137$, $\sin^2\theta_W = 0.2314$	≈ 0.630

Comparison

Both methods yield g_W in the range ≈ 0.63–0.65. The agreement at the $\sim$3% level confirms the internal consistency of the electroweak Standard Model at tree level. The small residual discrepancy is expected from higher-order radiative corrections.

4.1 Bringing the Methods into Exact Agreement

To show the methods agree exactly at tree level, note that the tree-level relations give:

$$m_W = \frac{1}{2} g_W v = \frac{g_e}{2\sin\theta_W} v = \frac{v}{2} \frac{\sqrt{4\pi\alpha}}{\sin\theta_W}$$

Closing the Loop: The On-Shell Scheme

If instead of using the $\overline{\text{MS}}$ value $\sin^2\theta_W = 0.2314$ (which is a running parameter evaluated at the Z-pole scale), we use the **on-shell definition**:

$$\sin^2\theta_W^{\text{on-shell}} \equiv 1 - \frac{m_W^2}{m_Z^2} = 1 - \frac{(80.379)^2}{(91.188)^2} = 1 - 0.7771 = 0.2229$$

then $\sin\theta_W^{\text{on-shell}} = 0.4721$, and:

$$g_W = \frac{g_e}{\sin\theta_W^{\text{on-shell}}} = \frac{0.3028}{0.4721} \approx 0.641$$

This is closer to the Part 1 value, though still not exact, because the relation $e = g_W \sin\theta_W$ also receives vertex and self-energy corrections at one loop. In the Standard Model with all radiative corrections included, all definitions of g_W converge to a single consistent value.

5 Bonus: Cross-Check via the Fermi Constant

For completeness, one can also extract g_W from the Fermi constant relation (Griffiths Eq. 9.36 in the revised edition):

The Fermi Constant Relation

$$\frac{G_F}{(\hbar c)^3} = \frac{\sqrt{2}}{8}\left(\frac{g_W}{M_W c^2}\right)^2$$

Solving for g_W:

$$g_W = M_W c^2 \sqrt{\frac{4\sqrt{2}\,G_F}{(\hbar c)^3}} = M_W c^2 \sqrt{4\sqrt{2}\,G_F}$$

(in natural units where $\hbar = c = 1$).

Using $G_F = 1.1664 \times 10^{-5}$ GeV^{-2} and $M_W = 80.379$ GeV:

$$g_W = 80.379 \times \sqrt{4\sqrt{2} \times 1.1664 \times 10^{-5}} \tag{5}$$

$$= 80.379 \times \sqrt{4 \times 1.4142 \times 1.1664 \times 10^{-5}} \tag{6}$$

$$= 80.379 \times \sqrt{6.596 \times 10^{-5}} \tag{7}$$

$$= 80.379 \times 8.122 \times 10^{-3} \tag{8}$$

Bonus Cross-Check Result

$$g_W \approx 0.653$$

This matches the Part 1 result exactly, as expected, because both the Fermi constant relation and the Higgs VEV relation are simply different ways of writing the *same* underlying physics (the W-boson propagator in the low-q^2 limit).

Summary of All Three Routes

Method	g_W
Higgs VEV: $g_W = 2m_W/v$	0.653
Electroweak mixing: $g_W = g_e/\sin\theta_W$	0.630
Fermi constant: $g_W = M_W\sqrt{4\sqrt{2}\,G_F}$	0.653

All three methods agree within a few percent — the residual spread is due to radiative corrections. The "$\alpha_W = g_W^2/(4\pi) \approx 1/29$" weak fine structure constant is *larger* than $\alpha_{\text{EM}} = 1/137$, showing that the weak force is intrinsically stronger than electromagnetism. The apparent "weakness" is entirely due to the massive W and Z propagators suppressing low-energy processes.